



2025 PRELIMINARY YEARLY EXAMINATION

Mathematics Extension

General Instructions

- Reading time – 10 minutes
- Working time – 2 hours
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided at the back of this paper
- In Questions 11 – 14, show relevant mathematical reasoning and/or calculations

Total marks:
70

Section I – 10 marks (pages 2 – 5)

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section

Section II – 60 marks (pages 6 – 10)

- Attempt Questions 11 – 14
- Allow about 1 hour and 45 minutes for this section

Section I

10 marks

Attempt Questions 1 – 10

Allow about 15 minutes for this section

Use the multiple-choice answer page in the writing booklet for Questions 1 – 10.

- 1 The polynomial $2x^3 + 6x^2 - 7x - 10$ has zeros α , β and γ . What is the value of $\alpha\beta\gamma(\alpha + \beta + \gamma)$?

A. -60
B. -15
C. 15
D. 60

- 2 Which of the following best describe the solution to the given equation $|x - 5| = 2x$?

A. 5
B. $\frac{5}{3}$
C. -5 and $\frac{5}{3}$
D. 5 and $\frac{5}{3}$

- 3 In how many ways can 7 people be chosen from a group of 16 people and then arranged in a circle?

A. $\frac{15!}{9!}$
B. $\frac{15!}{9!7!}$
C. $\frac{16!}{9!}$
D. $\frac{16!}{9!7}$

- 4 Which of the following is the coefficient of the fourth term in the expansion of $(3x - 4)^6$?
- A. -34560
B. 34560
C. 25920
D. -25920
-

- 5 Which of the following values of x could satisfy $2\sin^{-1}x + \cos^{-1}x = \frac{\pi}{3}$?
- A. $-\frac{1}{2}$
B. $\frac{1}{2}$
C. $\frac{\sqrt{3}}{2}$
D. $-\frac{\sqrt{3}}{2}$
-

- 6 What is the range of the function $y = \tan^{-1}(\sin x)$?
- A. $y \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$
B. $y \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$
C. $y \in \left[-\frac{\pi}{4}, \frac{\pi}{4}\right]$
D. $y \in \left(-\frac{\pi}{4}, \frac{\pi}{4}\right)$

7 Which of the following is an expression for $\frac{dy}{dx}$ if $x = 2\left(t + \frac{1}{t}\right)$ and $y = 2\left(t - \frac{1}{t}\right)$?

A. $\frac{t^2 + 1}{t^2 - 1}$

B. $\frac{t + 1}{t - 1}$

C. $\frac{t^2 - 1}{t^2 + 1}$

D. $\frac{t - 1}{t + 1}$

8 In triangle ABC, if $\cos A = \frac{1}{3}$, then which of the following equals to $\cot\left(\frac{A}{2}\right)$?

A. $\sqrt{2}$

B. $-\sqrt{2}$

C. $\frac{1}{2}$

D. $\frac{1}{\sqrt{2}}$

9 The polynomial $P(x) = x^3 + 5x^2 + x - 2$ has three real zeros. If $P(x)$ is translated to the right by 1 unit, what is the product of the zeros of the transformed polynomial?

A. -5

B. -1

C. 1

D. 5

- 10 The sum of the solutions of $\sin(2x) = \frac{\sqrt{3}}{2}$ over the interval $[-\pi, d]$ is $-\pi$. What could be the value of d ?

- A. 0
- B. $\frac{\pi}{6}$
- C. $\frac{\pi}{3}$
- D. $\frac{7\pi}{6}$

End of Section I

Section II

60 marks

Attempt Questions 11 – 14

Allow about 1 hour and 45 minutes for this section

Answer each question in the appropriate section of the writing booklet. Extra writing paper is available.

In Questions 11 – 14, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (16 marks)

- a) A scholarship committee needs to assign one of 4 ranks (Gold, Silver, Bronze, Participation) to each student. What is the minimum number of students required to ensure that at least one rank is given to 6 students? **1**

- b) Show that $\frac{\cos 23^\circ - \sqrt{3} \sin 23^\circ}{2} = \cos 83^\circ$. **2**

- c) The polynomial $P(x) = x^3 - 4x^2 + kx + 10$ has two zeros that are the same in magnitude but opposite in sign. Find the value of k and all zeros of the polynomial. **3**

- d) A curve is defined by the following parametric equations: **3**

$$x = -3 + 2 \cos^2 \theta \text{ and } y = -5 + \cos 2\theta$$

Find the Cartesian equation of the curve, noting any restrictions.

- e) If α, β and γ are roots of $2x^3 - 4x^2 + 1 = 0$, find the value of:

i) $\alpha^2 + \beta^2 + \gamma^2$ **2**

ii) $\frac{\alpha}{\beta\gamma} + \frac{\beta}{\alpha\gamma} + \frac{\gamma}{\alpha\beta}$ **1**

Question 11 continues next page

- f) Consider the function $f(x) = 2\cos^{-1}x$
- i) State the domain and range of $y = f(x)$. 2
- ii) Sketch the graph $y = |2\cos^{-1}x - \pi|$ 2

End of Question 11

Question 12 (15 marks)

- a) The polynomial $P(x)$ is given by $P(x) = x^3 - (k-1)x^2 + (1-k)x + 1$ for some real number k .
- i) Show that $x = -1$ is a root of the equation $P(x) = 0$. 1
- ii) Given that $P(x) = (x+1)(x^2 - kx + 1)$, find the set of values of k such that the equation $P(x) = 0$ has three distinct real roots. 3
- b) A bowl of hot soup at temperature $T^\circ\text{C}$, is placed in a cooler environment. It loses heat according to the law $\frac{dT}{dt} = k(T - T_0)$, where k is constant of proportionality, t is the time elapsed in minutes, and T_0 is the temperature of the environment in degrees Celsius.
- i) Show that $T = T_0 + Ae^{kt}$ satisfies the above equation, where A is constant. 1
- ii) A bowl of soup at 96°C is left to stand in a room at a temperature of 18°C . After 3 minutes the soup cools down to 75°C . Calculate the value of k correct to 4 decimal places. 2
- iii) Susan wishes to enjoy her soup at a temperature of 60°C . How long should she wait? 2
Answer to the nearest minutes.

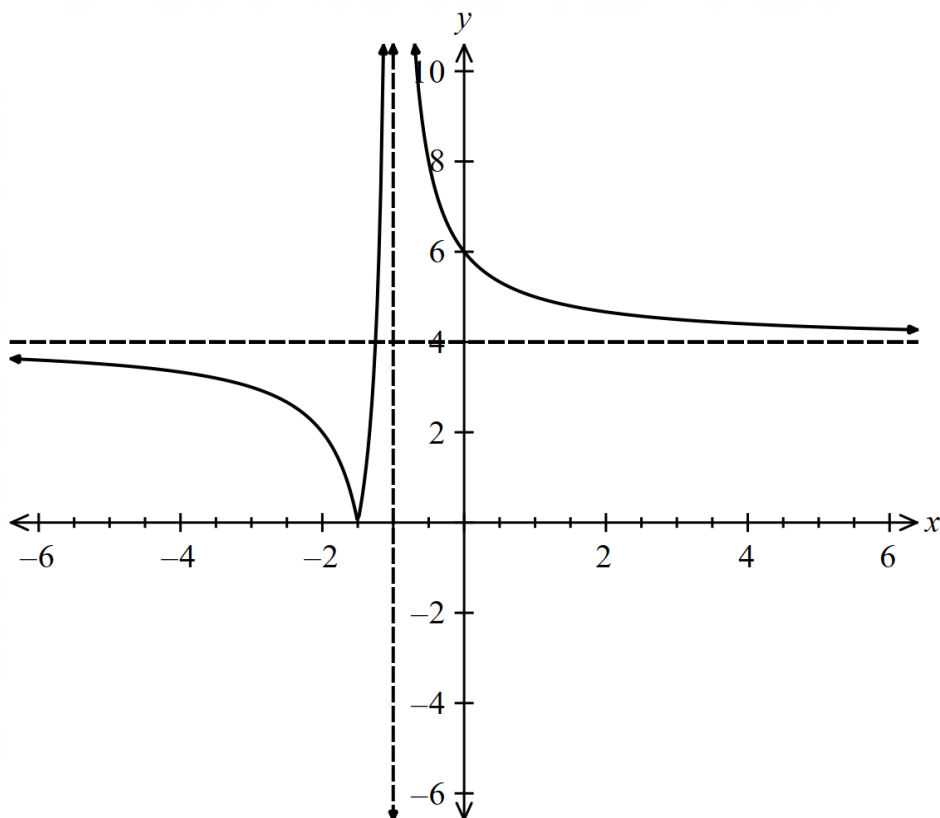
Question 12 continues next page

- c) A total of five players is selected at random from four sporting teams. Each of the teams consists of ten players numbered from 1 to 10.
- i) What is the probability that of the five selected players, three are numbered 4 and two are numbered 6? **2**
- ii) What is the probability that the five selected players contain at least four players from the same team? **2**
- d) Prove that $\tan^{-1} \frac{2}{3} + \cos^{-1} \frac{2}{\sqrt{5}} = \tan^{-1} \frac{7}{4}$. **2**

End of Question 12

Question 13 (15 marks)

- a) The diagram below is a sketch of the graph of the function $y = |f(x)|$, where $f(x) = \frac{2}{x+1} + 4$ and horizontal asymptote at $y = 4$.



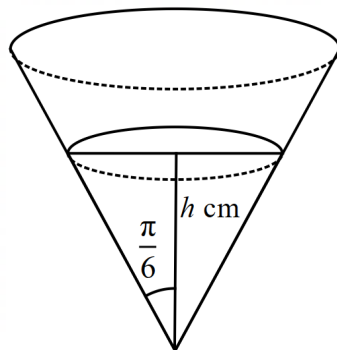
On separate diagrams, sketch the following graphs, in each case showing the intercepts on the axes and any asymptotes.

i) $|y| = f(|x|)$ 2

ii) $y = \sqrt{f(x)}$ 2

Question 13 continues next page

- b) An egg timer in the shape of an inverted right circular cone with semi-vertical angle $\frac{\pi}{6}$ radians. It contains sand to a depth of h cm as shown in the diagram below, which flows out of the apex of the cone at a constant rate of $0.5 \text{ cm}^3/\text{s}$.



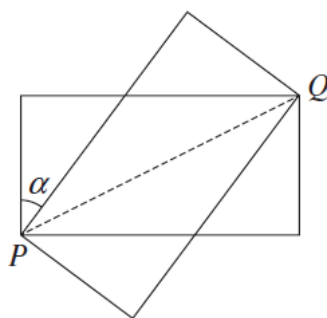
- i) Show that the volume $V \text{ cm}^3$ of sand in the cone is given by $V = \frac{1}{9} \pi h^3$ 1
- ii) Find the value of h when the depth of sand in the egg timer is decreasing at a rate of 0.05 cm/s , giving your answer correct to 2 decimal places. 3
- c) Given that $\sin x + \cos x = k$, find the value of $\sin(2x)$ in terms of k . 2
- d) Consider the function $f(x) = ax^3 + bx^2 + cx + d$, where a, b, c and d are real numbers.
- i) Write down the expression for $f'(x)$. 1
- ii) Given that the inverse of $f(x)$ is not a function, use part i) to show that $b^2 - 3ac > 0$. 2
- iii) Consider $g(x) = \frac{1}{2}x^3 - 3x^2 + 6x - 8$, use the result in part (ii) to show that the inverse of $g(x)$ is a function. 2

End of Question 13

Question 14 (14 marks)

a) Solve for x : $|x-1| > 2\sqrt{x(1-x)}$ **3**

- b) Two identical rectangles share a common diagonal PQ as shown in the following diagram. For each rectangle, the ratio of the longer side to the shorter side is $x : 1$. The angle at P between the sides of two rectangles is α . If $\alpha = \frac{\pi}{3}$ radians, then show that $x = \sqrt{3} + 2$. **3**



c) i) Show that $x^n (1+x)^n \left(1 + \frac{1}{x}\right)^n = (1+x)^{2n}$ **2**

ii) Hence show that $1 + \binom{n}{1}^2 + \binom{n}{2}^2 + \dots + \binom{n}{n}^2 = 2^n C_n$ **3**

d) Given that $\frac{\sin \beta}{\sin(2\alpha + \beta)} = \frac{m}{n}$, prove that $\frac{n-m}{n+m} \tan(\alpha + \beta) = \tan \alpha$. **3**

End of Exam

Preliminary Yearly Task-2025, Mathematics Extension-1

Marking Guidelines

Section - I

Multiple Choice (1 - 10)

- | | |
|------|-------|
| 1) B | 6) A |
| 2) B | 7) A |
| 3) D | 8) C |
| 4) A | 9) B |
| 5) A | 10) C |

Sample Solutions:

1) $2x^3 + 6x^2 - 7x - 10$ has zeros α, β and γ .

$$\therefore \alpha + \beta + \gamma = -\frac{6}{2} = -3 \quad \& \quad \alpha\beta\gamma = -\frac{(-10)}{2} = 5$$

$$\therefore \alpha\beta\gamma(\alpha + \beta + \gamma) = 5(-3)$$

$$= -15$$

$\therefore \textcircled{B}$.

2) $|x - 5| = 2x$; Condition is $x > 0$

$$x - 5 = \pm 2x$$

$$\therefore x \mp 2x = 5$$

$$-x = 5 \quad \text{or} \quad 3x = 5$$

$$x = -5 \quad \text{or} \quad x = \frac{5}{3}$$

Since $x > 0$, $\therefore x = \frac{5}{3}$ is only solution. $\therefore \textcircled{B}$

3) ${}^{16}C_7$ then arrange 7 people in circle

$$\therefore {}^{16}C_7 \cdot 6!$$

$$\frac{16!}{7! \cdot 9!} \times 6! = \frac{16!}{9! \times 7}$$

$\therefore \textcircled{D}$

4) $(3x - 4)^6$

General term: $T_{r+1} = {}^nC_r x^{n-r} y^r$ for the expansion of $(x+y)^n$

For fourth term sub $r=3$

$$\therefore T_4 = {}^6C_3 (3x)^3 (-4)^3$$

$\therefore \textcircled{A}$

Coefficient of 4th term: ${}^6C_3 (3)^3 (-4)^3 = -34560$.

5) $2\sin^{-1}x + \cos^{-1}x = \frac{\pi}{3}$

$$\sin^{-1}x + \frac{\pi}{2} = \frac{\pi}{3} \quad (\because \sin^{-1}x + \cos^{-1}x = \frac{\pi}{2})$$

$$\sin^{-1}x = \frac{\pi}{3} - \frac{\pi}{2}$$

$$= -\frac{\pi}{6}$$

$$x = \sin\left(-\frac{\pi}{6}\right)$$

$$x = -\sin\frac{\pi}{6} = -\frac{1}{2}$$

$\therefore \textcircled{A}$

$$6) \quad \cos A = \frac{1}{3}$$

$$\text{Let } t = \tan \frac{A}{2}$$

$$\frac{1-t^2}{1+t^2} = \frac{1}{3}$$

$$3 - 3t^2 = 1 + t^2$$

$$4t^2 = 2$$

$$t^2 = \frac{2}{4}$$

$$t = \frac{1}{\sqrt{2}} \quad (\because t > 0)$$

$$\therefore \tan \frac{A}{2} = \frac{1}{\sqrt{2}}$$

$$\cot \frac{A}{2} = \sqrt{2}$$

$\therefore \textcircled{A}$

$$7) \quad x = 2\left(t + \frac{1}{t}\right), \quad y = 2\left(t - \frac{1}{t}\right)$$

$$\frac{dx}{dt} = 2\left(1 - \frac{1}{t^2}\right)$$

$$\frac{dy}{dt} = 2\left(1 + \frac{1}{t^2}\right)$$

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$$

$$= \frac{2\left(1 + \frac{1}{t^2}\right)}{2\left(1 - \frac{1}{t^2}\right)}$$

$$= \frac{2(t^2+1)/t^2}{2(t^2-1)/t^2}$$

$$= \frac{t^2+1}{t^2-1}$$

$\therefore \textcircled{A}$

$$8) y = \tan^{-1}(\sin x)$$

$$\text{Since } -1 \leq \sin x \leq 1$$

$$\therefore \tan^{-1}(-1) \leq \tan^{-1}(\sin x) \leq \tan^{-1}(1)$$

$$-\frac{\pi}{4} \leq \tan^{-1}(\sin x) \leq \frac{\pi}{4}$$

$\therefore \textcircled{C}$

$$9) P(x) = x^3 + 5x^2 + x - 2$$

$x \rightarrow x-1$ as $P(x)$ translated 1 unit to the right.

$\therefore$ Translated Polynomial is

$$(x-1)^3 + 5(x-1)^2 + (x-1) - 2$$

$$x^3 - 3x^2 + 3x - 1 + 5(x^2 - 2x + 1) + (x-1) - 2$$

$$x^3 - 3x^2 \dots \dots \underbrace{-1 + 5(1) - 1 - 2}_{\text{constant term.}}$$

$\therefore$ Product of the zeros of translated poly. is

$$\frac{-\text{Constant term}}{\text{Coeff. of } x^3}$$

$$-\frac{1}{1} = -1$$

$\therefore \textcircled{B}$

$$10) \sin 2x = \frac{\sqrt{3}}{2} ; -\pi \leq x \leq d \quad \text{for } -2\pi \leq 2x \leq 2d$$

$$2x = -\pi - \frac{\pi}{3}, -2\pi + \frac{\pi}{3}, \frac{\pi}{3}, 2\frac{\pi}{3}, 2\pi + \frac{\pi}{3}, \dots$$

$$2x = -\frac{4\pi}{3}, -\frac{5\pi}{3}, \frac{\pi}{3}, 2\frac{\pi}{3}, \frac{7\pi}{3}, \dots$$

$$x = -\frac{2\pi}{3}, -\frac{5\pi}{6}, \frac{\pi}{6}, \frac{\pi}{3}, \frac{7\pi}{6}, \dots$$

Option A: Not possible as for $d=0$

$$\text{Sum of roots} = -\frac{2\pi}{3} + -\frac{5\pi}{6} = -\frac{3\pi}{2} \neq -\pi$$

Option B: Not possible as for $d = \frac{\pi}{6}$

$$\text{Sum of roots} = -\frac{2\pi}{3} - \frac{5\pi}{6} + \frac{\pi}{6} = -\frac{4\pi}{3} \neq -\pi$$

Option C: for $d = \frac{\pi}{3}$

$$\text{Sum of roots} = -\frac{2\pi}{3} - \frac{5\pi}{6} + \frac{\pi}{6} + \frac{\pi}{3} = -\pi$$

$\therefore \textcircled{C}$

END OF SECTION - 1

11 a) There are 4 ranks (Gold, Silver, Bronze, participation)

At least one rank is given to 6 students means

$$4 \times 5 + 1 = 21$$

1 correct solution

$\therefore$ Minimum number of students will be 21.

b) Show that $\frac{\cos 23^\circ - \sqrt{3} \sin 23^\circ}{2} = \cos 83^\circ$.

$$\text{R.H.S} = \cos 83^\circ$$

2 correct proof.

$$= \cos(60^\circ + 23^\circ)$$

1 correctly identify

$$= \cos 60^\circ \cos 23^\circ - \sin 60^\circ \sin 23^\circ$$

$$\cos 83^\circ = \cos(60^\circ + 23^\circ)$$

$$= \frac{1}{2} \cos 23^\circ - \frac{\sqrt{3}}{2} \sin 23^\circ$$

and apply $\cos(A+B)$

$$= \frac{\cos 23^\circ - \sqrt{3} \sin 23^\circ}{2}$$

$$= \text{L.H.S}$$

c) $P(x) = x^3 - 4x^2 + kx + 10$.

Roots are: $\alpha, -\alpha, \beta$

$$\sum \alpha = \alpha + (-\alpha) + \beta = 4 \Rightarrow \beta = 4 \quad \text{--- (1)}$$

$$\sum \alpha\beta = \alpha(-\alpha) - \alpha\beta + \alpha\beta = k$$

$$-\alpha^2 = k \quad \text{--- (2)}$$

$$\prod \alpha = \alpha(-\alpha)\beta = -10$$

$$-\alpha^2\beta = -10$$

$$4k = -10 \quad (\text{from (1) \& (2)})$$

$$k = -5/2$$

$$\therefore x^2 = \frac{5}{2}$$

$$x = \pm \sqrt{\frac{5}{2}}$$

$$\therefore \text{Roots are } \sqrt{\frac{5}{2}}, -\sqrt{\frac{5}{2}}, 4$$

$$\text{and } k = -\frac{5}{2}$$

3 Correct solution

2 Find correct value of k and one of the root.

1 Correctly apply sum or product of the roots.

$$11d) \quad x = -3 + 2\cos^2\theta \quad \& \quad y = -5 + \cos 2\theta$$

$$2\cos^2\theta = x + 3$$

$$\cos 2\theta = y + 5$$

$$\text{Since } \cos 2\theta = 2\cos^2\theta - 1$$

2 Correct solution

$$\therefore y + 5 = x + 3 - 1$$

$$y = x - 3$$

$$\text{Since } 0 \leq \cos^2\theta \leq 1$$

$$0 \leq 2\cos^2\theta \leq 2$$

$$\therefore -3 \leq -3 + 2\cos^2\theta \leq -1$$

$$-3 \leq x \leq -1$$

3 Correct solution

$$\therefore y = x - 3, \quad -3 \leq x \leq -1$$

2 Correct Cartesian equation

1 Eliminates parameter θ by using valid trig. identity.

11e) $2x^3 - 4x^2 + 1 = 0$

$$\sum \alpha = \frac{4}{2} = 2 \quad \sum \alpha\beta = 0 \quad \prod \alpha = -\frac{1}{2}$$

i) $\alpha^2 + \beta^2 + \gamma^2 = (\sum \alpha)^2 - 2(\sum \alpha\beta)$ 2 correct solution

$$= \left(\frac{4}{2}\right)^2 - 2(0)$$

$$= 4$$

1 Uses $\alpha^2 + \beta^2 + \gamma^2 = (\sum \alpha)^2 - 2(\sum \alpha\beta)$

ii) $\frac{\alpha}{\beta\gamma} + \frac{\beta}{\alpha\gamma} + \frac{\gamma}{\alpha\beta}$

$$\frac{\alpha^2 + \beta^2 + \gamma^2}{\alpha\beta\gamma} = \frac{(\sum \alpha)^2 - 2(\sum \alpha\beta)}{\alpha\beta\gamma}$$

$$= \frac{(2)^2 - 0}{-\frac{1}{2}}$$

$$= \frac{4}{-\frac{1}{2}} = -8$$

1 correct solution

11f)

i) $f(x) = 2\cos^{-1}x$

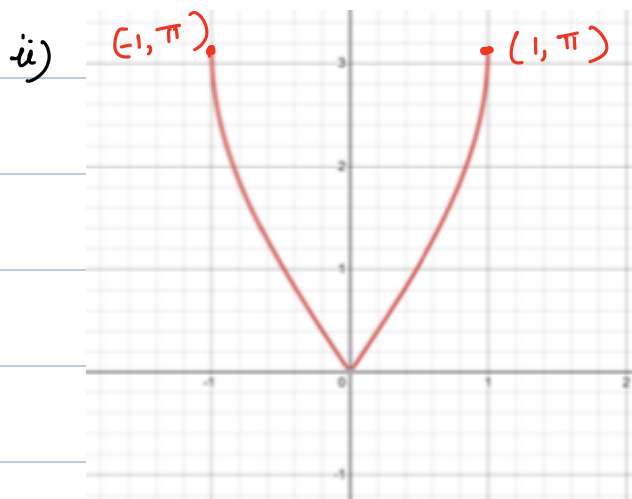
2 marks Correct Solution

Domain: $\{x : -1 \leq x \leq 1\}$

1 mark Correct domain or

Range: $\{y : 0 \leq y \leq 2\pi\}$

range of $f(x)$.



2 marks Correct sketch

1 mark Correct sketch for $2\cos^{-1}x$

or Correct sketch for $2\cos^{-1}x - \pi$

or Correct sketch for $|2\cos^{-1}x|$

Note: Sketch should have correct

shape including endpoints and cannot look "smooth" at the origin

12(a)

$$i) P(-1) = (-1)^3 - (K-1)(-1)^2 + (1-K)(-1) + 1$$

$$= -1 - (K-1) - (1-K) + 1$$

1 mark Correct Solution.

$$= -1 - K + 1 - 1 + K + 1$$

shows $P(-1) = 0$ by substitution.

$$= 0$$

ii) Since $P(x) = 0$ has 3 real roots

3 marks Correct Solution

$\therefore x^2 - Kx + 1 = 0$ has 2 real roots

2 marks Finds discriminant

$$\therefore \Delta > 0$$

of quadratic in terms of K

$$K^2 - 4 > 0$$

and realise $\Delta > 0$ for real roots

$$(K-2)(K+2) > 0$$

1 mark Deduce $x^2 + Kx + 1 = 0$

$$\therefore K < -2 \text{ or } K > 2.$$

has real roots.

12 b)

$$i) T = T_0 + Ae^{kt} \quad \text{--- (1)}$$

1 mark Correct solution.

$$\frac{dT}{dt} = \frac{d}{dt}(T_0 + Ae^{kt})$$

$$= kAe^{kt}$$

$$= k(T - T_0) \quad [\text{from (1) } T - T_0 = Ae^{kt}]$$

$\therefore$ (1) satisfies the given differential equation.

ii) At $t=0$, $T=96$, $T_0=18$

$$\therefore 96 = 18 + A \Rightarrow A = 78$$

$$\therefore T = 18 + 78e^{kt}$$

When $t=3$, $T=75^\circ$

$$\therefore 75 = 18 + 78e^{3k}$$

$$e^{3k} = \frac{57}{78}$$

$$3k = \ln \frac{57}{78}$$

$$k = -0.1046$$

2 Correct Solution.

1 find A or

Sub $t=3$, $T=75^\circ$ in

$$T = T_0 + Ae^{kt}$$

iii) $t = ?$ $T = 60^\circ$

$$T = 18 + 78e^{kt}$$

$$60 = 18 + 78e^{kt}$$

$$e^{kt} = \frac{42}{78}$$

$$kt = \ln \frac{42}{78}$$

$$t = \frac{1}{k} \ln \frac{42}{78} \quad \text{where } k = -0.1046$$

$$\approx 5.92 \text{ minutes.}$$

6 minutes

2 correct solution

1 Sub $T=60^\circ\text{C}$ in

$T = T_0 + Ae^{kt}$ to find e^{kt} .

12c)

i) There are 4 players numbered '6' and 4 players numbered '8' from a total of forty players.

Three '6's can be selected in 4C_3 ways

Two '8's can be selected in 4C_2 ways

Five players can be selected in ${}^{40}C_5$ ways.

$$\therefore \text{Probability} = \frac{{}^4C_3 \times {}^4C_2}{{}^{40}C_5}$$

$$= \frac{4 \times 6}{658008}$$

$$= \frac{1}{27417}$$

2 correct solution.

1 Correctly find total
 ${}^{40}C_5$ (or) ${}^4C_3 \times {}^4C_2$

(or) equivalent
approach.

ii) 'At least 4 players' means 4 or 5 players.

5 players from one team can be selected in ${}^{10}C_5$ ways.

Since there are 4 teams, therefore 5 players from the same team can be selected in $({}^4C_1 \times {}^{10}C_5)$ ways

4 players from one team and one player from the remaining teams (30 players) can be selected in $({}^4C_1 \times {}^{10}C_4 \times {}^{30}C_1)$

$$\therefore \text{Probability} = \frac{({}^4C_1 \times {}^{10}C_5) + ({}^4C_1 \times {}^{10}C_4 \times {}^{30}C_1)}{{}^{40}C_5}$$

$$= \frac{28}{703}$$

2 Correct Solution.

1 Correctly find ${}^4C_1 \times {}^{10}C_5$ (or)
 ${}^4C_1 \times {}^{10}C_4 \times {}^{30}C_1$ (or) equivalent approach

12d)

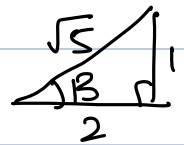
Prove $\tan^{-1} \frac{2}{3} + \cos^{-1} \frac{2}{\sqrt{5}} = \tan^{-1} \frac{7}{4}$

or $\tan\left(\tan^{-1} \frac{2}{3} + \cos^{-1} \frac{2}{\sqrt{5}}\right) = \frac{7}{4}$

Let $\tan^{-1} \frac{2}{3} = A$ and $\cos^{-1} \frac{2}{\sqrt{5}} = B$

$$\tan A = \frac{2}{3}$$

$$\cos B = \frac{2}{\sqrt{5}}$$



$$\tan B = \frac{1}{2}$$

$$\text{L.H.S} = \tan(A+B)$$

$$= \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

$$= \frac{\frac{2}{3} + \frac{1}{2}}{1 - \frac{2}{3} \times \frac{1}{2}}$$

$$= \frac{7/6}{4/6}$$

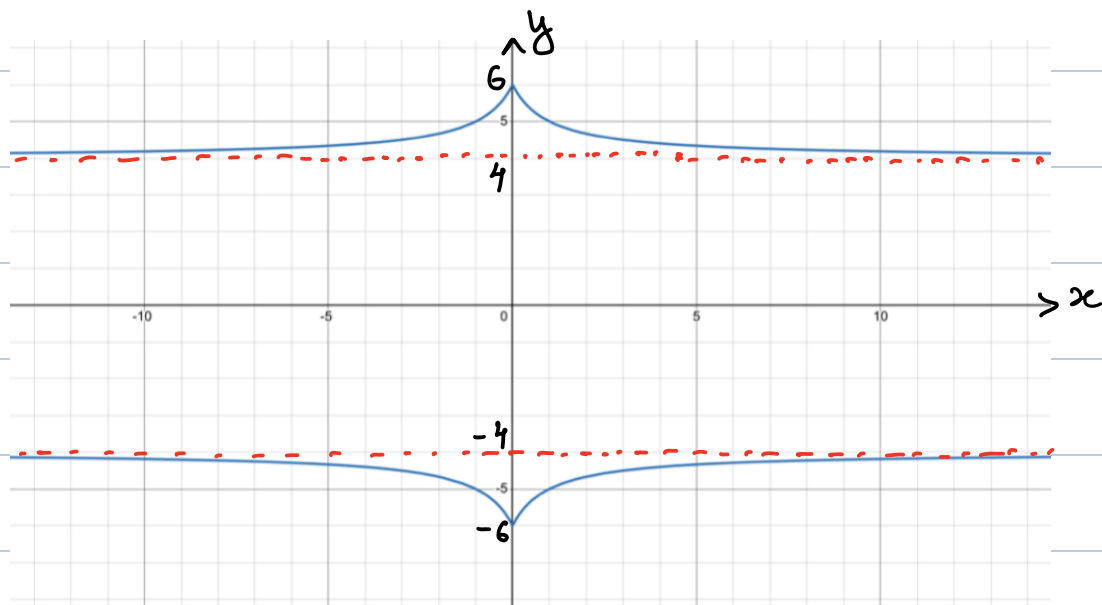
$$= \frac{7}{4}$$

$$= \text{R.H.S}$$

2 Correct proof

1 Obtaining $\tan B = \frac{1}{2}$ or equivalent approach.

13 a)i)

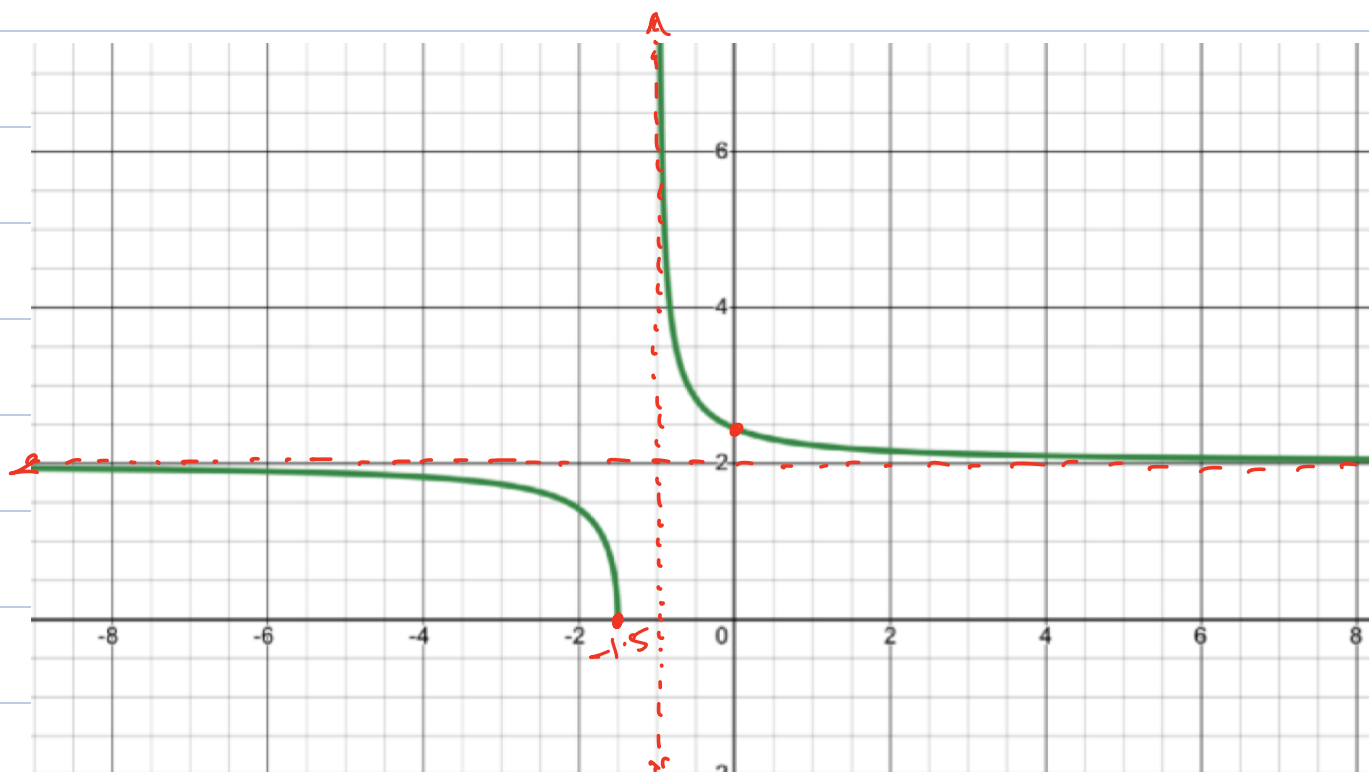


2 marks Correct sketch including shape and asymptotes.

1 mark One correct graph including shape & asymptotes

or both correct graph ($|y|$ & $f(x)$) without asymptotes

ii)



2 mark Correct graph including asymptotes.

1 mark Correct graph without asymptotes

(or) Correct graph for any one branch (either $x > -1$ or $x < -1$) including shape and x -intercept.

13 b)

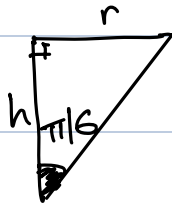
i) $V = \frac{1}{3} \pi r^2 h$

$$r = h \tan \frac{\pi}{6}$$

$$= \frac{h}{\sqrt{3}}$$

$$\therefore V = \frac{1}{3} \pi \left(\frac{h}{\sqrt{3}} \right)^2 h$$

$$= \frac{1}{9} \pi h^3$$



1 mark Correct Solution.

Express radius of the cone in terms of h and show substitution.

ii) $\frac{dh}{dt} = -0.05 \text{ cm/s}$, $h = ?$

$$\frac{dV}{dt} = -0.5 \text{ cm}^3/\text{s}$$

$$\frac{dV}{dt} = \frac{dV}{dh} \times \frac{dh}{dt}$$

$$-0.5 = \frac{1}{3} \pi h^2 \times (-0.05)$$

$$h^2 = \frac{30}{\pi}$$

$$h = 3.09 \text{ cm}$$

3 marks Correct Solution.

2 marks Express the chain rule in terms of h .

1 mark Obtains relevant chain rule 'or' finds $\frac{dV}{dh}$

13c) $\sin x + \cos x = k$

Square both sides

$$(\sin x + \cos x)^2 = k^2$$

$$\sin^2 x + \cos^2 x + 2\sin x \cos x = k^2$$

$$1 + 2\sin x \cos x = k^2 \quad (\because \sin^2 x + \cos^2 x = 1)$$

$$2\sin x \cos x = k^2 - 1 \quad \text{2 Correct Solution}$$

$$\sin 2x = k^2 - 1 \quad \text{1 Square both sides for}$$

$$\sin x + \cos x = k \text{ (or)}$$

Correct use of trig. Identities

13d) $f(x) = ax^3 + bx^2 + cx + d$.

i) $f'(x) = 3ax^2 + 2bx + c$

1 mark Correct solution

ii) If the inverse of $f(x)$ is not a function, then it is neither monotonically increasing / decreasing.

Since $f'(x)$ is a quadratic function and we don't want $f'(x) \geq 0$ or $f'(x) \leq 0$ for all value of x .

Therefore for $f'(x)$, Set $\Delta > 0$.

$$f'(x) = 3ax^2 + 2bx + c$$

2 marks Correct Solution

$$\Delta = (2b)^2 - 4(3a)(c) > 0$$

1 mark finds discriminant for $f'(x)$ (or)

$$4b^2 - 12ac > 0$$

$$\text{or } b^2 - 3ac > 0$$

Explains why $\Delta > 0$ for $f'(x)$.

Hence proved.

iii) Compare the coefficient of $g(x)$ and $f(x)$.

$$a = \frac{1}{2}, \quad b = -3, \quad c = 6, \quad d = -8.$$

For $g^{-1}(x)$ to exist; $b^2 - 3ac = 0$, which shows only one x -intercept.

$$\begin{aligned} \therefore b^2 - 3ac &= (-3)^2 - 3\left(\frac{1}{2}\right)(6) \\ &= 9 - 9 \\ &= 0 \end{aligned}$$

Since $b^2 - 3ac = 0$, therefore from ii) inverse of $g(x)$ is a function.

2 marks Correct solution

1 mark Evaluate $b^2 - 3ac$

14a)

$$|x-1| > 2\sqrt{x(1-x)}$$

$\sqrt{x(1-x)}$ is defined for $0 \leq x \leq 1$

$$|x-1|^2 > (2\sqrt{x(1-x)})^2$$

$$(x-1)^2 > 4x(1-x)$$

$$x^2 - 2x + 1 > 4x - 4x^2$$

$$5x^2 - 6x + 1 > 0$$

$$5x^2 - 5x - x + 1 > 0$$

$$5x(x-1) - (x-1) > 0$$

$$(5x-1)(x-1) > 0$$

$$\therefore x < \frac{1}{5} \text{ or } x > 1$$

Since function is defined for $0 \leq x \leq 1$

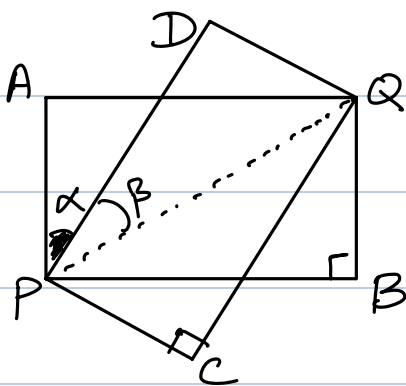
$$\therefore 0 \leq x < \frac{1}{5}$$

3 marks - correct solution (graphical/algebraic)

2 marks - Correctly solve the quadratic inequality.

1 mark - Identify the function is defined as $0 \leq x \leq 1$, or
Square both sides to find correct quad. eq.

14b)



Let $\angle QPD = \beta$

In $\triangle PDQ$; $\tan \beta = \frac{1}{x}$

In $\triangle APQ$; $\tan(\alpha + \beta) = x$

$$\therefore \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta} = x$$

$$\frac{\tan \frac{\pi}{3} + \frac{1}{x}}{1 - \tan \frac{\pi}{3} \cdot \frac{1}{x}} = x \quad \left(\because \alpha = \frac{\pi}{3} \text{ \& } \tan \beta = \frac{1}{x} \right)$$

$$\frac{\sqrt{3} + \frac{1}{x}}{1 - \frac{\sqrt{3}}{x}} = x$$

$$\frac{\sqrt{3}x + 1}{x - \sqrt{3}} = x$$

$$\sqrt{3}x + 1 = x^2 - \sqrt{3}x$$

$$x^2 - 2\sqrt{3}x - 1 = 0$$

$$x = \frac{2\sqrt{3} \pm \sqrt{12 + 4}}{2}$$

$$= \frac{2\sqrt{3} \pm 4}{2}$$

$$= \sqrt{3} \pm 2$$

$$= \sqrt{3} + 2 \quad (\because x > 0)$$

Hence Shown

3 correct solution

2 Use Compound angle

Formula for $\tan(\alpha + \beta)$

Or, equivalent approach

1 Find $\tan \beta = \frac{1}{x}$ or

correct approach to

find relation between angles

$$\begin{aligned}
 14c) \text{ L.H.S} &= x^n (1+x)^n \left(1+\frac{1}{x}\right)^n \\
 &= x^n (1+x)^n \left(\frac{x+1}{x}\right)^n \\
 &= \frac{x^n (1+x)^n (x+1)^n}{x^n} \\
 &= (1+x)^n (x+1)^n \\
 &= [(1+x)^n]^2 \\
 &= (1+x)^{2n} \\
 &= \text{R.H.S}
 \end{aligned}$$

Hence Shown.

2 Correct proof

1 Express $\left(1+\frac{1}{x}\right)^n$ as $\left(\frac{x+1}{x}\right)^n$

'or' $x^n \left(1+\frac{1}{x}\right)^n = (1+x)^n$

'or' equivalent approach.

ii) Compare the coefficient of x^n for part i)

Coefficient of x^n in the expansion of $(1+x)^{2n}$ is $^{2n}C_n$.

$$\text{L.H.S} = x^n (1+x)^n \left(1 + \frac{1}{x}\right)^n$$

$$= x^n \left(1 + {}^nC_1 x + {}^nC_2 x^2 + \dots + {}^nC_n x^n\right) \left[1 + {}^nC_1 \left(\frac{1}{x}\right) + {}^nC_2 \left(\frac{1}{x}\right)^2 + \dots + {}^nC_n \left(\frac{1}{x}\right)^n\right]$$

for the coefficient of x^n in this expansion

$$x^n (\text{Const. term}) (\text{Const. term}) = x^n \cdot 1 \cdot 1 = 1$$

$$x^n \cdot {}^nC_1 x \cdot {}^nC_1 \left(\frac{1}{x}\right) = {}^nC_1 \cdot {}^nC_1 = ({}^nC_1)^2$$

$$x^n \cdot {}^nC_2 x^2 \cdot {}^nC_2 \left(\frac{1}{x}\right)^2 = {}^nC_2 \cdot {}^nC_2 = ({}^nC_2)^2$$

$$\vdots \quad \vdots \quad \vdots \quad \vdots \quad \vdots$$

$$x^n \cdot {}^nC_n x^n \cdot {}^nC_n \left(\frac{1}{x}\right)^n = {}^nC_n \cdot {}^nC_n = ({}^nC_n)^2$$

$\therefore$ Coefficient of x^n in the expansion of L.H.S is

$$1 + ({}^nC_1)^2 + ({}^nC_2)^2 + \dots + ({}^nC_n)^2$$

By comparing the coefficient of x^n of both sides of part i)

we get;

$$1 + ({}^nC_1)^2 + ({}^nC_2)^2 + \dots + ({}^nC_n)^2 = {}^{2n}C_n$$

Hence shown.

3 correct proof

2 Correct expansion of coefficient of x^n for
and $(1+x)^{2n}$
 $x^n (1+x)^n \left(1 + \frac{1}{x}\right)^n$ by using binomial expansion

1 correct expansion of $(1+x)^{2n}$ by using binomial
expansion or finding the coefficient of x^n for
 $(1+x)^{2n}$

$$14 d) \frac{\sin \beta}{\sin(2\alpha + \beta)} = \frac{m}{n} \quad \text{--- (1)}$$

Prove that $\frac{n-m}{n+m} \tan(\alpha + \beta) = \tan \alpha$

$$R.H.S = \frac{n-m}{n+m} \tan(\alpha + \beta)$$

$$= \frac{1 - \frac{m}{n}}{1 + \frac{m}{n}} \tan(\alpha + \beta)$$

$$= \frac{1 - \frac{\sin \beta}{\sin(2\alpha + \beta)}}{1 + \frac{\sin \beta}{\sin(2\alpha + \beta)}} \tan(\alpha + \beta) \quad \text{--- (using (1))}$$

$$= \frac{\sin(2\alpha + \beta) - \sin \beta}{\sin(2\alpha + \beta) + \sin \beta} \cdot \tan(\alpha + \beta)$$

$$= \frac{2 \cos\left(\frac{2\alpha + \beta + \beta}{2}\right) \sin\left(\frac{2\alpha + \beta - \beta}{2}\right)}{2 \sin\left(\frac{2\alpha + \beta + \beta}{2}\right) \cos\left(\frac{2\alpha + \beta - \beta}{2}\right)} \cdot \tan(\alpha + \beta)$$

$$= \frac{\cos(\alpha + \beta) \sin \alpha}{\sin(\alpha + \beta) \cos \alpha} \cdot \frac{\sin(\alpha + \beta)}{\cos(\alpha + \beta)}$$

$$= \frac{\sin \alpha}{\cos \alpha}$$

$$= \tan \alpha$$

$$= L.H.S$$

3 correct proof.

2 Simplify $\sin(2\alpha + \beta)$ and $\tan(\alpha + \beta)$ by using correct trig. Identity (OR) use of product to sum results.

1 Eliminating m and n .

END OF EXAM

—x—x—